Predicting Movie Ratings Based on Metadata

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***Abstract*—Accurate prediction of movie ratings from publicly available metadata can significantly aid decision-making in the film industry. In this work, we compile a dataset of 7,036 films (1975–2024) collected from IMDb, Kaggle, and GitHub. Features are 23 attributes. Such as, Runtime, release year, budget adjust by inflation, popularity, facebook actor director likely, genre, and etc. After rigorous cleaning, we are using imputation, removal outlier, normalization, and encoding technique to transform data.**

We tested multiple MLP architectures (from shallow to deep) and found the best single hidden layer model—a 64-neuron layer with *tanh* activation, trained for 100 epochs with batch size 64—which achieves MSE = 0.704 and R^2 = 0.384 on a held-out test set. We also compared against a Random Forest baseline, which yields MSE = 0.6186 and R^2 = 0.4589 on its test split, with 5-fold cross-validation reporting MSE = 0.6832 $\pm$ 0.0476 and R^2 = 0.4256 $\pm$ 0.0253.

While the ANN provides valuable insights into hyperparameter impacts. However the random tree-based methods demonstrate superior predictive power on tabular movie metadata. Our findings offer practical guidelines for selecting and tuning regression models in entertainment analytics, balancing architectural complexity against generalization ability.

Keywords: Movie Rating Prediction, Deep Learning, Regression, MLP-ANN, Random Forest, IMDb, Kaggle, Github

I. I.Introduction

The prediction of movie ratings has become an increasingly important task in the era of digital entertainment, where user-generated reviews and ratings play a critical role in shaping audience preferences, marketing strategies, and content production decisions. Accurate rating prediction models can benefit a wide range of stakeholders, including film studios, streaming platforms, and recommendation systems, by enabling more informed decisions regarding movie production, acquisition, and promotion.

Movie ratings, such as those provided by IMDb, are influenced by a multitude of factors, including cast, crew, budget, genre, language, and release timing. The complexity of these relationships, combined with the heterogeneity and i ncompleteness of real-world data, presents significant challenges for predictive modeling. Traditional approaches, such as collaborative filtering and content-based methods, often fall short in capturing the nuanced interactions among diverse metadata features, especially for movies that have not yet been released or lack sufficient historical data.

Recent advances in machine learning, particularly in feature engineering and deep learning, offer new opportunities to address these challenges. By integrating and preprocessing data from multiple sources—including IMDb, Kaggle, and external budget datasets—and engineering features that reflect the impact of actors, directors, genres, and financial investment, it is possible to build more robust and accurate predictive models. In this project, we develop a comprehensive machine learning pipeline that leverages both traditional regression techniques and artificial neural networks to predict IMDb movie ratings prior to release. Our approach emphasizes rigorous data cleaning, inflation adjustment of financial features, and the use of advanced feature engineering to maximize predictive performance and interpretability.

The remainder of this paper is organized as follows: Section II describes the problem statments and explain why this research is important; Section III presents exploratory data analysis and feature selection; Section IV details the modeling approach and evaluation metrics; Section V discusses experimental results and comparative analysis; and Section VI concludes with insights and directions for future work.

II. Problem Statement

Accurately predicting movie ratings before a film's release is a significant challenge in the entertainment industry, with direct implications for studios, streaming platforms, and marketing strategies. Movie ratings, such as those provided by IMDb, are influenced by a complex interplay of factors including cast, crew, budget, genre, and release timing. Traditional approaches often rely on limited features or historical performance, failing to capture the nuanced relationships present in modern, data-rich environments.

The primary objective of this project is to develop a robust machine learning framework capable of predicting IMDb movie ratings using a diverse set of metadata features available prior to release. This involves integrating and cleaning data from multiple sources (IMDb, Kaggle, and external budget datasets), engineering features that reflect the impact of actors, directors, genres, and financial investment, and addressing challenges such as missing values, data heterogeneity, and feature selection. By leveraging advanced regression models, including deep neural networks, we aim to improve the accuracy and interpretability of movie rating predictions, providing valuable insights for stakeholders in the film industry.

III. Motivation

The ability to accurately predict movie ratings prior to a film's release holds significant value for a variety of stakeholders in the entertainment industry. For film studios and producers, reliable rating predictions can inform investment decisions, marketing strategies, and content development, ultimately reducing financial risk and increasing the likelihood of commercial success. Streaming platforms and recommendation systems can leverage such predictions to enhance user experience by curating content that aligns with audience preferences and expectations.

From a research perspective, movie rating prediction presents a complex and multifaceted challenge. Ratings are influenced by a diverse set of factors, including cast, crew, budget, genre, and release timing, as well as external elements such as market trends and audience sentiment. The integration of heterogeneous data sources, the need for advanced feature engineering, and the application of state-of-the-art machine learning models make this problem both technically demanding and intellectually stimulating.

Furthermore, the growing availability of large-scale movie metadata and user-generated ratings provides an unprecedented opportunity to develop and evaluate predictive models that can generalize across different genres, time periods, and production contexts. By addressing the challenges inherent in movie rating prediction, this research contributes to the broader fields of data science, machine learning, and entertainment analytics, offering insights and methodologies that can be applied to other domains characterized by complex, high-dimensional data.

IV. Dataset

The dataset used in this study is constructed by merging multiple sources to provide a comprehensive set of features for movie rating prediction. The primary data sources include the IMDb Non-Commercial Datasets[1], the Kaggle metadata, and external budget data from Kaggle[2]. The key files and their contents are as follows:

- **IMDb:**
  - `title.ratings.tsv.gz`: Contains average ratings (`averageRating`) and number of votes for each movie.
  - `title.basics.tsv.gz`: Includes genres, release year (`startYear`), runtime (`runtimeMinutes`), and primary title information.
  - `title.crew.tsv.gz`: Provides director and writer IDs.
  - `title.principals.tsv.gz`: Lists main cast and crew IDs, focusing on the top actors.
- **Kaggle:** Provides additional metadata such as budget, revenue, and production companies.
- **Kaggle Budget Dataset:** Used to supplement missing budget information in IMDB and Kaggle.

After merging and cleaning, the final dataset contains over 7,000 movies with features including genres, cast, crew, budget (inflation-adjusted), runtime, language, and release year. Figure 1 shows the distribution of IMDb ratings, and Figure 2 visualizes the distribution of (log-transformed) movie budgets.

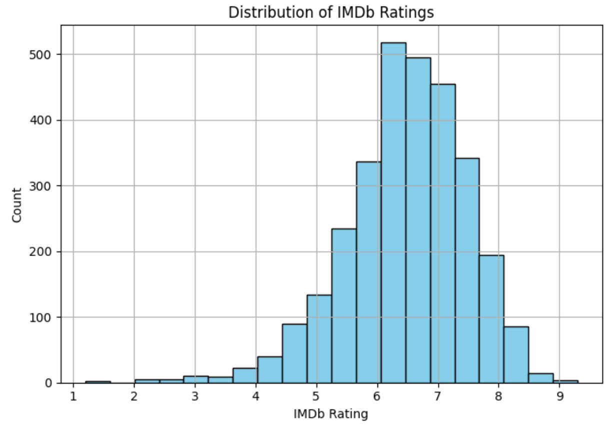

Fig. 1. Distribution of IMDb Ratings

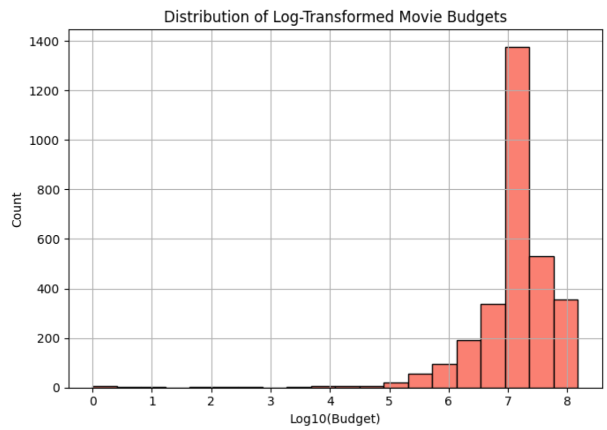

Fig. 2. Distribution of Log-Transformed Movie Budgets

V. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution and relationships of key features in the dataset, guiding feature selection and preprocessing decisions. The following visualizations and analyses were performed:

- **Distribution of IMDb Ratings:** Figure 1 shows that movie ratings are approximately normally distributed, with a slight skew towards higher ratings. Most movies have ratings between 5 and 8, with fewer movies at the extremes.
- **Distribution of Movie Budgets:** Figure 2 presents the log-transformed budget distribution, which normalizes the strong right-skewness observed in raw budget values. The majority of movies have budgets between 1 million and 100 million USD.
- **Correlation Heatmap:** Figure **??** visualizes the correlations between numeric features such as budget, runtime, popularity, and average rating. Notably, budget and popularity show moderate positive correlation with ratings, while runtime has a weaker relationship.
- **Genre Distribution:** Figure 3 displays the frequency of

different genres in the dataset. Drama, comedy, and action are the most common genres, while niche genres are less represented.

These analyses informed the selection of features for modeling and highlighted the need for normalization and careful handling of outliers. The visualizations also revealed the importance of genre and budget in predicting movie ratings.

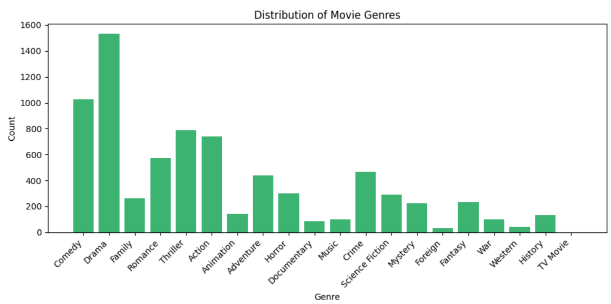

Fig. 3. Distribution of Movie Genres

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